



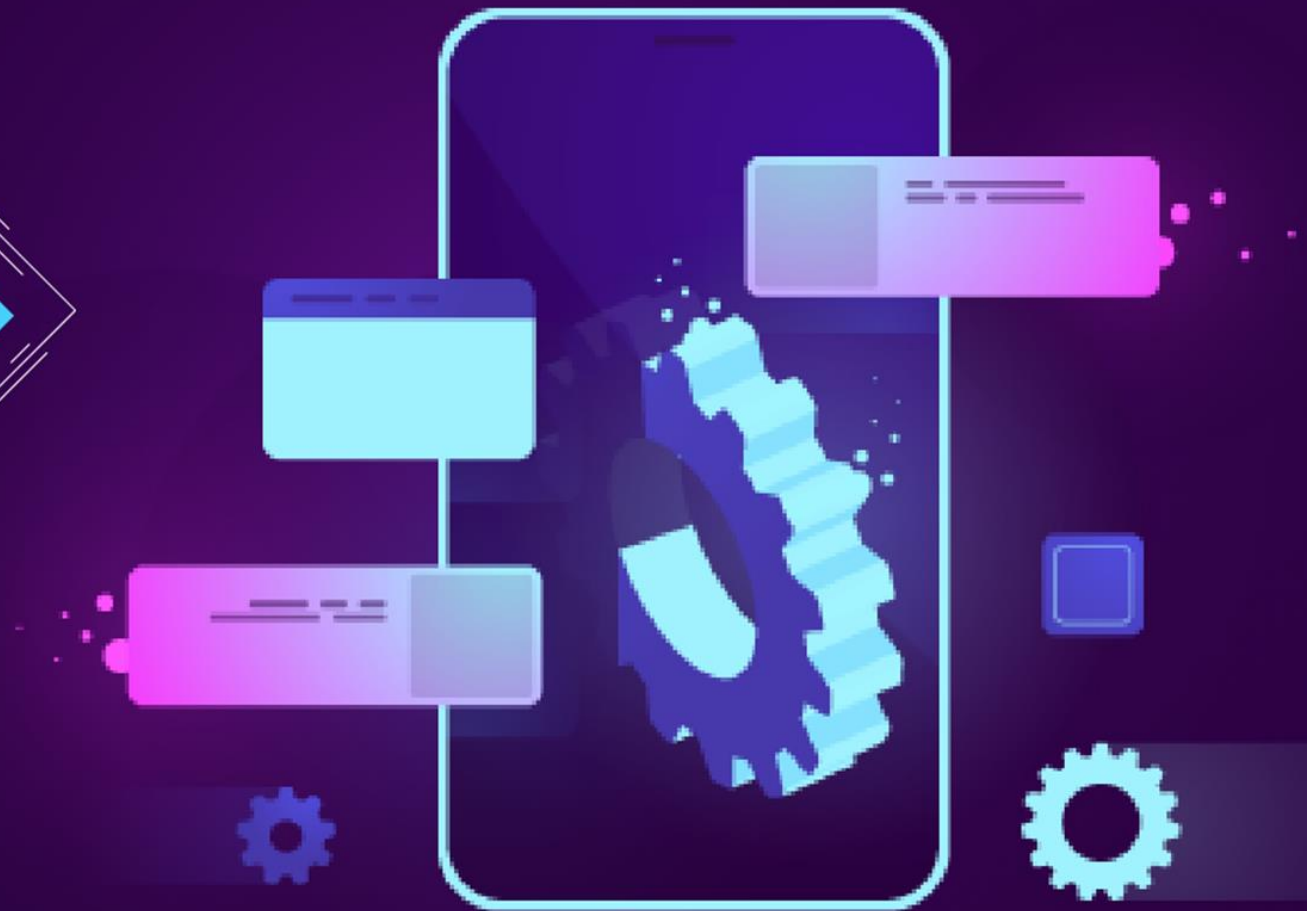
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Flutter Developer & Instructor



Flutter 3



Session – 17 : State Management in Flutter





$$\text{UI} = f(\text{state})$$

The layout
on the screen

Your
build
methods

The application state

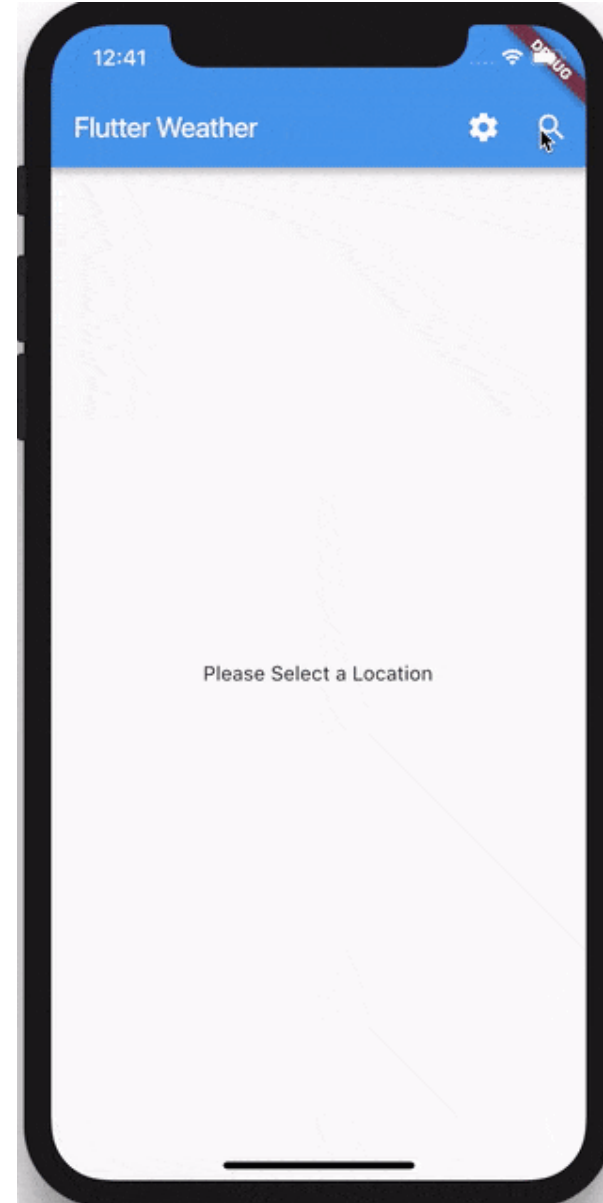
What is a State



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HANDLING STATES



```
class CounterState {}  
  
class CounterInitial extends CounterState {}  
  
class UpdateCounter extends CounterState {}
```



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HANDLING CUBIT



```
import 'package:first_app/cubit/counter_state.dart';
import 'package:flutter_bloc/flutter_bloc.dart';

class CounterCubit extends Cubit<CounterState> {
  CounterCubit() : super(CounterInitial());

  int counter = 0;

  void add() {
    counter++;
    emit(UpdateCounter());
  }

  void remove() {
    counter--;
    emit(UpdateCounter());
  }
}
```



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BLOC PROVIDER

is used as a dependency injection (DI) widget so that a single instance of a bloc can be provided to multiple widgets within a subtree.

```
Widget build(BuildContext context) {  
  return BlocProvider(  
    create: (context) => CounterCubit(),  
    child: //UI of Screen  
  );  
}
```



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BLOC BUILDER

handles building the widget in response to new states. Use **BlocListener** if you want to "do" anything in response to state changes such as navigation, showing a dialog, etc...

```
Widget build(BuildContext context) {  
  return BlocProvider(  
    create: (context) => CounterCubit(),  
    child: BlocBuilder<CounterCubit, CounterState>(  
      builder: (context, state) {  
        return Scaffold(  
          // UI OF SCREEN  
        );  
      },  
    ),  
  );  
}
```



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BLOC LISTENER

It should be used for functionality that needs to occur once per state change such as navigation, showing a SnackBar, showing a Dialog, etc...

```
Widget build(BuildContext context) {  
  return BlocProvider(  
    create: (context) => CounterCubit(),  
    child: BlocProvider(  
      create: (context) => CounterCubit(),  
      child: BlocListener<CounterCubit, CounterState>(  
        listener: (context, state) {  
          // "do" anything in response to state changes such  
          // as navigation, showing a dialog  
        },  
        child:  
          // return UI of Screen  
      ),  
    ),  
  );  
}
```



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BLOC CONSUMER

It should only be used when it is necessary to both rebuild UI and execute other reactions to state changes in the bloc such as navigation, showing a dialog, etc...

```
Widget build(BuildContext context) {  
  return BlocProvider(  
    create: (context) => CounterCubit(),  
    child: BlocProvider(  
      create: (context) => CounterCubit(),  
      child: BlocConsumer<CounterCubit, CounterState>(  
        listener: (context, state) {  
          // "do" anything in response to state changes such  
          // as navigation, showing a dialog  
        },  
        builder: (context, state) {  
          // return UI of Screen  
        },  
      )),  
  );  
}
```



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DEALING WITH INSTANCE OF CUBIT



```
// with extensions  
context.read<CounterCubit>().add();  
  
// without extensions  
BlocProvider.of<CounterCubit>(context).add();
```



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